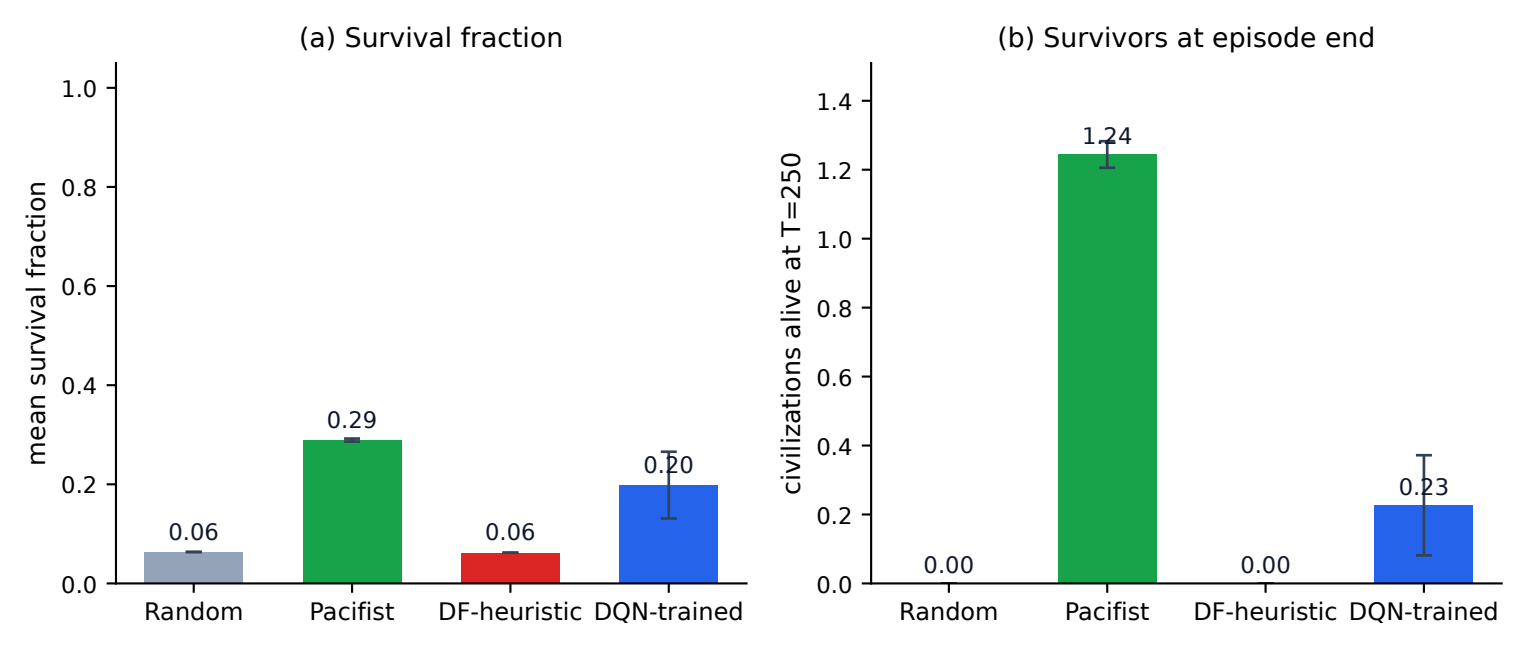
(a) Survival fraction
mean survival fraction
1.0
0.8
0.6
0.4
0.2
0.0
0.06
0.29
0.06
0.20
Random
Pacifist
DF-heuristic
DQN-trained
(b) Survivors at episode end
civilizations alive at T=250
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1.24
0.00
0.00
0.23
Random
Pacifist
DF-heuristic
DQN-trained